

Rotation angles of a rotating disc as the holonomy of the Hopf fibration

Takuya Matsumoto

Department of Applied Physics, Faculty of Engineering, University of Fukui, 3-9-1 Bunkyo, Fukui-shi, Fukui 910-8507, Japan

Department of Fundamental Engineering for Knowledge-Based Society, Graduate School of Engineering, University of Fukui, 3-9-1 Bunkyo, Fukui-shi, Fukui 910-8507, Japan

E-mail: takuyama@u-fukui.ac.jp

Abstract

We address the fundamental question of how rotation angles arise in classical rigid body motion from a geometric viewpoint. As a fundamental kinematical setting, we consider a rotating-disc model. We show that the rotation angle accumulated in a cyclic motion can be naturally decomposed into a dynamical phase and a geometric phase, and that the latter is obtained from the $U(1)$ holonomy of the Hopf fibration equipped with its canonical connection through a natural weight-two representation of the fiber group. By introducing a Gauss map based on a normal vector fixed at the center of the rotating disc, the physical periodic motion is mapped to a closed curve on the two-sphere S^2 , whose horizontal lift in the Hopf bundle $S^3 \rightarrow S^2$ determines the geometric phase through the weight-two representation. This framework provides a conceptually clear interpretation of rotation angles as geometric phases and clarifies the geometric origin of their decomposition.

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1 Introduction and summary

Rotation angles arising in simple mechanical systems often exhibit features that cannot be fully explained by purely local or dynamical considerations. A classical example is provided by a disc rolling without slipping along a prescribed path, where the total rotation accumulated after a closed cycle generally differs from what would be expected from naive kinematical arguments. Such phenomena are naturally described in terms of a decomposition of the total rotation angle into a dynamical contribution and an additional phase of geometric origin. Similar decompositions appear in a wide range of contexts, from rigid body motion to geometric phases in classical and quantum mechanics.

The purpose of the present paper is not to introduce a new mechanical model, but to clarify the geometric nature of this additional rotation in a particularly simple and concrete setting. We show that the geometric contribution to the rotation angle of a disc rolling along the rim of a fixed disc admits a natural interpretation as the holonomy associated with parallel transport in a principal $U(1)$ bundle. More precisely, by representing the kinematics of the rolling disc through an appropriate Gauss map, the motion is mapped to a closed curve on the two-sphere S^2 , and the accumulated rotation is obtained from the $U(1)$ holonomy of the Hopf fibration $S^3 \rightarrow S^2$ endowed with its canonical connection through the weight-two representation. In this way, a familiar mechanical system provides a concrete realization of a fundamental concept in differential geometry.

This article provides a detailed mathematical exploration of a simple kinematical model, called “*Rotation angles of a rotating disc*,” in order to elucidate the concept of geometric phase, a universal and interdisciplinary phenomenon in physics [1, 2, 3, 4, 5, 6]. The primary goal is to interpret the geometric contribution to the rotation angle in terms of the holonomy of the Hopf fibration and its natural weight-two representation. This clarifies the geometric significance of its components. Hence, our model is a typical model that exhibits the geometric phase in a simple setup.

The physical model The setup involves two discs (Disc A and B) in a three-dimensional space $\mathbb{R}^3$. Disc A is a fixed disc with radius a , centered at the origin on the $z = 0$ plane (the xy -plane), while Disc B with radius b is another disc that rolls on the edge of Disc A without slipping. The position of Disc B is determined by two parameters: the angle θ , which indicates the point of contact on Disc A’s circumference, and the angle β between Disc B and xy -plane. The motion is described over a time interval $t \in [0, 1]$, during which Disc B starts from an initial configuration and returns to the same position at $t = 1$.

The central question and decomposition of rotation angle The core question this paper seeks to answer is as follows: “How much does the disc B rotate in this motion from $t = 0$ to $t = 1$?” The total rotation angle, denoted as Δ , is shown to be decomposable into two distinct parts, the *dynamical phase* (Δ_d) and *geometric phase* (Δ_g). The former, Δ_d , depends on the ratio of the radii of Discs A and B; for a one-cycle motion, $\Delta_d = 2\pi a/b$. The latter, Δ_g , depends only on the geometric motion of Disc B as a rigid body in $\mathbb{R}^3$. The article presents explicit formulas for both phases, but notes that the geometric meaning of the formula (2.4) for Δ_g is initially opaque and requires further elaboration.

Geometric interpretation via Gauss map To unravel the geometric meaning, this study introduces a *Gauss map*. This map associates the orientation of Disc B with a unit normal vector, $g(\theta, \beta) \in S^2 \subset \mathbb{R}^3$. As Disc B completes its cyclic motion, the tip of this vector traces an oriented closed curve γ (the *Gauss curve*) on the unit two-sphere (S^2). The geometric phase Δ_g is then expressed in terms of geometric properties of this curve γ . When the Gauss curve is simple, the geometric phase Δ_g is expressed in terms of geometric properties of γ by

$$\Delta_g = A_+ - 2\pi I_+, \quad (1.1)$$

where A_+ is the area enclosed by the curve on the sphere, and I_+ is a *topological number* representing the number of poles enclosed by $\gamma(\epsilon)$, a regularized version of the curve.

The Hopf fibration and holonomy The main theoretical development of the paper compared with our previous work [8] is to connect this geometric phase to the underlying

mathematical structure, the *Hopf fibration*. The Hopf fibration describes the three-sphere (S^3) as a principal fiber bundle over S^2 with a $U(1)$ fiber. The article demonstrates that the physical geometric phase is obtained from the $U(1)$ *holonomy* $\tau(\gamma)$ of the Hopf fibration associated with the Gauss curve γ , through the natural weight-two representation of the fiber group. The key ingredient is the *canonical connection* ω in Def.-Prop. 3.4, which determines the horizontal lift of the Gauss curve. The main theorem of this study is as follows.

Main Theorem (Thm. 4.5). *Let γ be a piecewise smooth Gauss curve on $\mathbb{CP}^1$ associated with the motion (M). Let*

$$\rho_2 : U(1) \longrightarrow U(1), \quad \rho_2(u) = u^2, \quad (1.2)$$

be the weight-two representation of the fiber group. Then the geometric phase Δ_g and the holonomy $\tau(\gamma)$ of the Hopf fibration are related by

$$\rho_2(\tau(\gamma)) = \exp(i\Delta_g). \quad (1.3)$$

Thus, the physical geometric phase is obtained from the $U(1)$ holonomy of the Hopf fibration through the natural weight-two representation ρ_2 . This relation provides a precise geometric interpretation of the rotation angle that is not apparent from the original kinematical formula (2.4).

On the other hand, the $U(1)$ relation (1.3) loses the real-valued information contained in the geometric phase, including the topological contribution appearing in (1.1). This issue is addressed in § 4.2 by considering a real lift of the $U(1)$ holonomy. Let $p : \mathbb{R} \rightarrow U(1)$, $p(x) = e^{ix}$ be the universal covering map, and let

$$\tilde{\rho}_2 : \mathbb{R} \longrightarrow \mathbb{R}, \quad \tilde{\rho}_2(x) = 2x, \quad (1.4)$$

be the lift of the weight-two representation ρ_2 . For the representative of the lifted holonomy $\tilde{\tau}_M(\gamma)$ associated with the motion (M), we obtain $\tilde{\tau}_M(\gamma) = \pi n + \Delta_g/2$. Consequently,

$$\tilde{\rho}_2(\tilde{\tau}_M(\gamma)) = 2\pi n + \Delta_g. \quad (1.5)$$

These relations are summarized by the commutative diagram

$$\begin{array}{ccc} \mathbb{R} \ni \tilde{\tau}_M(\gamma) & \xrightarrow[\text{(1.4)}]{\tilde{\rho}_2} & \mathbb{R} \ni 2\pi n + \Delta_g \\ p \downarrow & & \downarrow p \\ U(1) \ni \tau(\gamma) & \xrightarrow[\text{(1.2)}]{\rho_2} & U(1) \ni \exp(i\Delta_g). \end{array} \quad (1.6)$$

Thus, the passage from the Hopf holonomy to the physical geometric phase can be described consistently both at the $U(1)$ level and at the level of its universal covering space.

A complementary geometric interpretation is developed in § 4.3. Let ϕ denote the angle of the tangent vector of the Gauss curve relative to a local moving frame on S^2 , and let κ_g be its geodesic curvature. Then, the geometric phase $\Delta_g(t)$ for any $t \in [0, 1]$ is determined by the horizontal condition for the canonical connection and given by (Prop. 4.10)

$$\Delta_g(t) = \phi(t) - \phi(0) - \int_0^{s(t)} \kappa_g(s) ds, \quad (1.7)$$

where s is the length parameter. Hence, the geometric phase can also be understood directly on the two-sphere as a combination of the net rotation of the tangent direction relative to the local frame and the accumulated geodesic curvature.

Contents The remainder of this paper is organized as follows. In § 2, we set up the model called *Rotation angles of a rotating disc*. § 3 is devoted to collecting the basic materials on the Hopf fibration, including the definition (§ 3.1), the canonical connection (§ 3.2), the decomposition of S^3 into the horizontal and vertical subspaces (§ 3.3), and the notion of parallel displacement (§ 3.4). In § 4, we demonstrate that the geometric phase of the rotation angle can be obtained from the $U(1)$ holonomy of the Hopf fibration through the weight-two representation (§ 4.1). We then introduce a real lift of the $U(1)$ holonomy and clarify how the real-valued geometric phase, including its topological information, is retained at the level of the universal covering space $\mathbb{R}$ (§ 4.2). Finally, we relate the canonical Hopf connection to the local moving frame and the geodesic curvature of the Gauss curve, providing a complementary geometric interpretation of the geometric phase directly on S^2 (§ 4.3). In App. A, we explicitly present a diffeomorphism of S^2 and $\mathbb{CP}^1$.

2 The model: Rotation angles of a rotating disc

In this section, we introduce a simple kinematical model¹, which we refer to as “*Rotation angles of a rotating disc*,” and summarize some of the results obtained for the model in [8].

¹This model was initially proposed as one of the common problems for the 32nd Japan Mathematics Contest and the 25th Japan Junior Mathematics Contest [7].

Rotation angles of a rotating disc

Consider a disc A of radius $a > 0$ lying in $z = 0$ plane (*i.e.* xy -plane) with its center at the origin in three-dimensional space $\mathbb{R}^3$. Set another disc B in $\mathbb{R}^3$ with the radius $b > 0$ so that it contacts disc A at $(a \cos \theta, a \sin \theta, 0) \in \mathbb{R}^3$ with $\theta \in \mathbb{R}$. Suppose that the line perpendicular to disc B and passing through its center is either parallel to the z -axis or intersects the z -axis. Denote the angle between $z = 0$ plane and the disc B by β^a , and suppose that $0 \leq \beta \leq \pi$, see Fig.1. The position of disc B is determined entirely by two parameters, θ and β .

Let disc B roll along the edge of disc A without slipping.

Introduce the time parameter $t \in [0, 1] \subset \mathbb{R}$ to describe this motion, and regard the angle variables θ and β as continuous functions of t . Then, the position of the rigid body, disc B, is determined by the map

$$(\theta, \beta) : [0, 1] \rightarrow \mathbb{R} \times [0, \pi], \quad t \mapsto (\theta(t), \beta(t)). \quad (\text{M})$$

Set $\theta(0) = 0$. Starting at $t = 0$, disc B returns to its initial position at $t = 1$. That is

$$\theta(1) = 2\pi n \quad \text{with} \quad n \in \mathbb{Z}, \quad \beta(1) = \beta(0). \quad (\text{T})$$

We shall call the integer $\theta(1)/2\pi = n$ the *topological number*.

^aComparing with our previous work [8], the definition of β is changed as $\beta_{[\text{here}]} = \pi - \beta_{[8]}$.

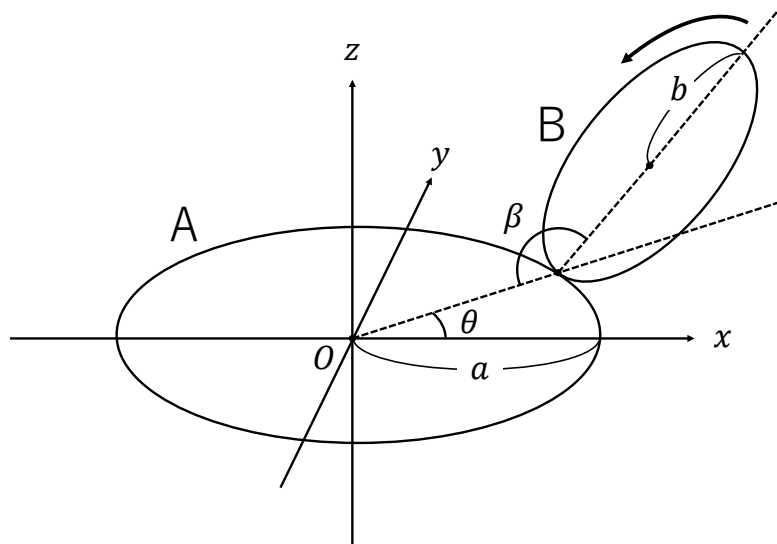


Figure 1: Disc A is fixed on xy -plane and disc B rolls on the edge of the disc A without slipping.

Remark 2.1. The functions $\theta(t)$ and $\beta(t)$ are assumed to satisfy the following conditions.

1. The continuous functions $\theta(t), \beta(t)$ satisfy the *Lipschitz condition*. That is, for any $t_1, t_2 \in [0, 1]$, there exists a positive constant C_f such that

$$|f(t_1) - f(t_2)| \leq C_f |t_1 - t_2|, \quad (2.1)$$

where f is either θ or β .

2. Both $\theta(t)$ and $\beta(t)$ are differentiable except at finitely many points on $(0, 1)$. The derivatives $\theta'(t), \beta'(t)$ are piecewise continuous on $[0, 1]$.

These two conditions are physically natural. Since $t \in [0, 1]$ is an auxiliary parameter to describe the motion (M), it is always possible to assume that $\theta(t), \beta(t)$ satisfy the above conditions.

Our interest lies in the following question.

Question

How much does the disc B rotate in this motion from $t = 0$ to $t = 1$? Define and determine the rotation angle Δ .

The answer to this question was provided by [8].

Definition-Theorem 2.2 ([8]). *The total rotation angle Δ of disc B is decomposed into the sum of the dynamical phase Δ_d and the geometric phase Δ_g ,*

$$\Delta = \Delta_d + \Delta_g, \quad (2.2)$$

and they are respectively given by

$$\Delta_d = \frac{2\pi na}{b} \quad (\text{Dynamical phase}), \quad (2.3)$$

$$\Delta_g = - \int_0^1 \cos \beta(t) \frac{d\theta(t)}{dt} dt \quad (\text{Geometric phase}). \quad (2.4)$$

Though the above theorem gives us an explicit formula for the geometric phase, the geometric meaning is opaque. We elaborate on this point in the following subsections.

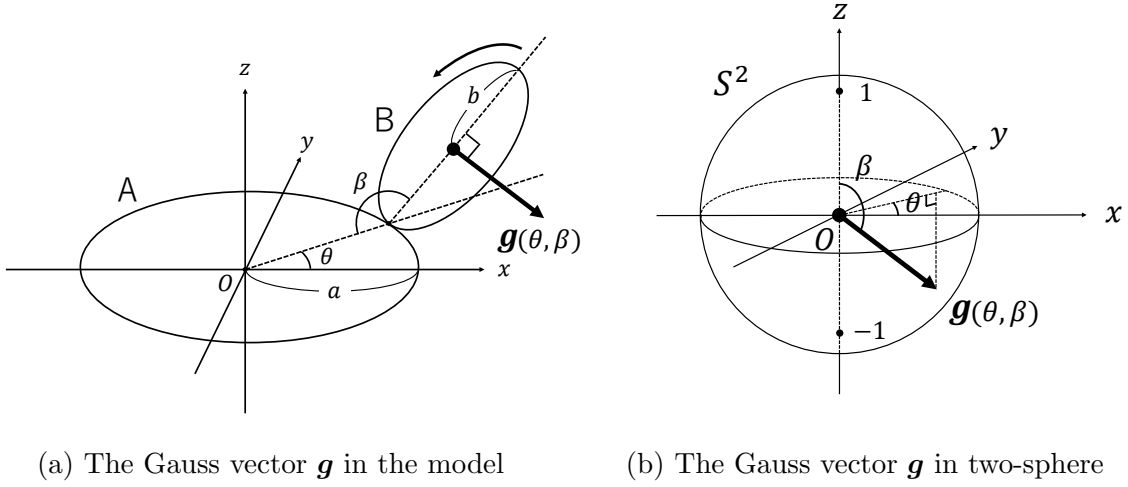


Figure 2: The Gauss map $\mathbf{g}$ from the model (a) to two-sphere (b)

Gauss map

First, we introduce the unit normal vector of disc B by *the Gauss map*, which detects the motion of disc B. We assume that two-sphere S^2 is embedded in $\mathbb{R}^3$ as

$$S^2 = \{ x^2 + y^2 + z^2 = 1 \mid \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \} \subset \mathbb{R}^3.$$

Definition 2.3 (Gauss map). For the two parameters (θ, β) in the motion (M), define *the Gauss map* from $\mathbb{R} \times [0, \pi]$ to two-sphere S^2 by

$$\mathbf{g} : \mathbb{R} \times [0, \pi] \rightarrow S^2 \subset \mathbb{R}^3, \quad (\theta, \beta) \mapsto \mathbf{g}(\theta, \beta) = \begin{pmatrix} \sin \beta \cos \theta \\ \sin \beta \sin \theta \\ \cos \beta \end{pmatrix}. \quad (2.5)$$

The vector $\mathbf{g}(\theta, \beta)$ is called *the Gauss vector*.

The Gauss vector is geometrically realized both in the rotating model and two-sphere as in Fig. 2a and Fig. 2b, respectively.

Note that $\mathbf{g}(\theta, \beta)$ is a unit vector and perpendicular to disc B. By composing the motion (M) and the Gauss map (2.5), the Gauss vector could be regarded as a function of $t \in [0, 1]$,

$$\mathbf{g} : [0, 1] \rightarrow S^2, \quad t \mapsto \mathbf{g}(t) = \begin{pmatrix} \sin \beta(t) \cos \theta(t) \\ \sin \beta(t) \sin \theta(t) \\ \cos \beta(t) \end{pmatrix}. \quad (2.6)$$

We also refer to the map in (2.6) as the Gauss map.

Because of $\mathbf{g}(1) = \mathbf{g}(0)$, the Gauss map defines the oriented closed curve γ on S^2 associated with the given motion (M) by

$$\gamma = \{ \mathbf{g}(t) \in S^2 \mid t \in [0, 1] \} \subset S^2. \quad (2.7)$$

The orientation of γ is induced from that of the parameter $t \in [0, 1]$ such as from $t = 0$ to $t = 1$. The curve γ is said to be *simple* if it does not intersect with itself.

The regularized curve

Second, though the Gauss map (2.6) detects the motion of the rotating disc, it does not move when the disc is in the xy plane, which corresponds to $\beta = 0, \pi$. To resolve this incompleteness, we introduce the following regularization. For a sufficiently small $\epsilon \in (0, \pi/8)$, which plays the role of *cut-off*, define the regularized angular coordinate by

$$\begin{aligned} \beta_\epsilon : [0, 1] &\rightarrow [\epsilon, \pi - \epsilon], \quad t \mapsto \beta_\epsilon(t), \\ \beta_\epsilon(t) &= \begin{cases} \epsilon & (0 \leq \beta(t) \leq \epsilon) \\ \beta(t) & (\epsilon < \beta(t) < \pi - \epsilon) \\ \pi - \epsilon & (\pi - \epsilon \leq \beta(t) \leq \pi) \end{cases} \end{aligned} \quad (2.8)$$

The regularized Gauss vector is then given by replacing $\beta(t)$ by $\beta_\epsilon(t)$ as

$$\mathbf{g}_\epsilon : [0, 1] \rightarrow S^2 - \{0, 0, \pm 1\}, \quad t \mapsto \mathbf{g}_\epsilon(t) = \begin{pmatrix} \sin \beta_\epsilon(t) \cos \theta(t) \\ \sin \beta_\epsilon(t) \sin \theta(t) \\ \cos \beta_\epsilon(t) \end{pmatrix}. \quad (2.9)$$

Definition 2.4 (Regularized Gauss curve). The *regularized Gauss curve* is defined by the orbit of the regularized Gauss vector,

$$\gamma(\epsilon) = \{ \mathbf{g}_\epsilon(t) \in S^2 - \{(0, 0, \pm 1)\} \mid t \in [0, 1] \} \subset S^2 - \{(0, 0, \pm 1)\}. \quad (2.10)$$

The point is that the curve $\gamma(\epsilon)$ dodges the poles $(0, 0, \pm 1)$, though the undeformed curve γ does not in general. By taking $\epsilon \rightarrow 0$, it continuously reduces to the original Gauss curve,

$$\gamma(\epsilon) \rightarrow \gamma. \quad (2.11)$$

The significance of $\gamma(\epsilon)$ is that it correctly reflects the topological indices $I_\pm$ as we will see later.² Adopting this regularization, we could rewrite the geometric phase (2.4) in terms of the line integral, as follows:

²For the details, see Remark 3.7 in [8].

Proposition 2.5 ([8]). *The geometric phase for the motion (M) is given by the limit of the line integral along the regularized curve $\gamma(\epsilon)$ in (2.10),*

$$\Delta_g = -\lim_{\epsilon \rightarrow 0} \int_{\gamma(\epsilon)} \cos \beta_\epsilon d\theta \quad (2.12)$$

Furthermore, the geometric meaning of the above line integral has been revealed in [8]. Denote by A_+ (A_-) the area enclosed by the Gauss curve γ on the left (right), and by I_+ (I_-) $\in \{0, 1, 2\}$ the total number of poles at $(0, 0, \pm 1) \in S^2$ enclosed by the regularized Gauss curve $\gamma(\epsilon)$ on the left (right, respectively). By the definitions, they satisfy

$$A_+ + A_- = 4\pi \quad \text{and} \quad I_+ + I_- = 2. \quad (2.13)$$

With these notations, the geometric phase Δ_g is expressed as follows:

Theorem 2.6 (Theorem 3.15 in [8]). *Suppose that the oriented closed curve γ defined in (2.7) is simple. Then, for the motion (M), the geometric phase is given by*

$$\Delta_g = A_+ - 2\pi I_+ = -A_- + 2\pi I_- = \frac{A_+ - A_-}{2} - \pi(I_+ - I_-). \quad (2.14)$$

Together with the dynamical phase Δ_d , the answer to this question is summarized as follows.

Answer for the Question

The rotation angle Δ of the rotating disc B in the motion (M) with the topological condition (T) is given by the sum of the dynamical phase Δ_d and the geometric phase Δ_g as follows,

$$\Delta = \Delta_d + \Delta_g \quad \text{with} \quad \Delta_d = \frac{2\pi na}{b}, \quad \Delta_g = A_+ - 2\pi I_+. \quad (2.15)$$

Here, A_+ is the area of the region surrounded by the curve γ in (2.7) which is supposed to be simple, and $I_+ \in \{0, 1, 2\}$ is the number of poles enclosed by the regularized curve $\gamma(\epsilon)$ in (2.10) on the left side. See also Fig. 3.

3 Hopf fibration

In this section, we shall review the definition of the Hopf fibration in § 3.1 and fix some conventions for subsequent arguments. In particular, the *canonical connection* in § 3.2 is important because it will be used to introduce the *horizontal* and *vertical* subspaces in § 3.3, and define the parallel displacement in § 3.4. The main references in this section are [9, 10, 11, 12].

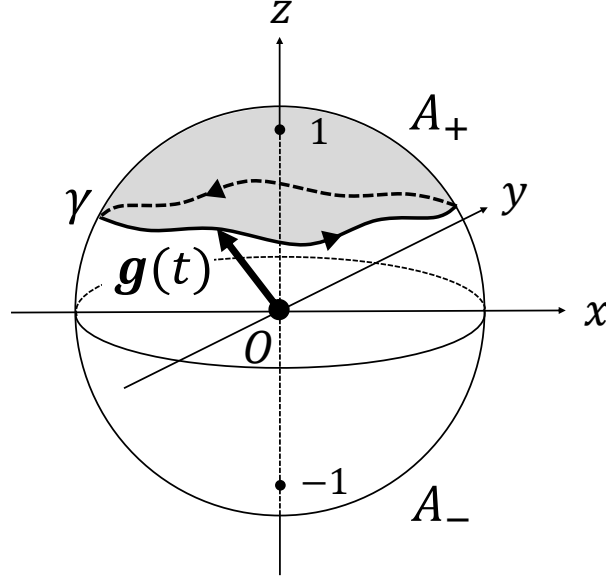


Figure 3: The curve $\gamma \subset S^2$ is the orbit of the Gauss vector $\mathbf{g}(t)$ whose orientation is induced by the parameter t . $A_+(A_-)$ is the area enclosed by γ on the left (right) side of the figure. The topological index $I_+(I_-)$ is the number of poles at $(0, 0, \pm 1) \in S^2$ contained in γ on the left (right) side. The above picture corresponds to the case $I_+ = I_- = 1$.

3.1 Definition of the Hopf fibration

The *Hopf fibration* [12] is a principal fiber bundle equipped with $U(1)$ fiber embedded in the total space S^3 and the projection called the *Hopf map* $\pi : S^3 \rightarrow S^2$, *i.e.*,

$$S^1 \hookrightarrow S^3 \xrightarrow{\pi} S^2. \quad (3.1)$$

To be more precise, define the three-sphere S^3 by

$$S^3 = \{ (z_1, z_2) \in \mathbb{C}^2 \mid |z_1|^2 + |z_2|^2 = 1 \} \subset \mathbb{C}^2. \quad (3.2)$$

The one-dimensional complex projective space $\mathbb{CP}^1$ is a quotient space of $\mathbb{C}^2 \setminus \{(0, 0)\}$ by $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$ action. Namely, the equivalence relation $(z_1, z_2) \sim (w_1, w_2)$ is defined by

$$(z_1, z_2) = (\lambda w_1, \lambda w_2) \quad \text{with} \quad \lambda \in \mathbb{C}^\times. \quad (3.3)$$

We denote a representative of $\mathbb{CP}^1$ by $[z_1, z_2]$, that is,

$$\mathbb{CP}^1 = \{ [z_1, z_2] \mid (z_1, z_2) \in \mathbb{C}^2 \setminus \{(0, 0)\} \}. \quad (3.4)$$

The projective space $\mathbb{CP}^1$ has an atlas $\{(U_1, f_1), (U_2, f_2)\}$ defined by

$$\begin{aligned} f_1 : U_1 &= \{ [z_1, z_2] \mid z_1 \neq 0 \} \rightarrow \mathbb{C}, & f_1([z_1, z_2]) &= \frac{z_2}{z_1}, \\ f_2 : U_2 &= \{ [z_1, z_2] \mid z_2 \neq 0 \} \rightarrow \mathbb{C}, & f_2([z_1, z_2]) &= \frac{z_1}{z_2}. \end{aligned} \quad (3.5)$$

Since $f_1(U_1 \cap U_2) = f_2(U_1 \cap U_2) = \mathbb{C}^\times$, the transition map reads

$$f_2 \circ f_1^{-1} : \mathbb{C}^\times \rightarrow \mathbb{C}^\times, \quad z \mapsto \frac{1}{z}. \quad (3.6)$$

Note that $\mathbb{CP}^1$ and S^2 are diffeomorphic and the explicit correspondence is summarized in App. A.

Definition 3.1 (Hopf fibration). The *Hopf fibration* $(S^3, \mathbb{CP}^1, \pi, U(1))$ is a principal fiber bundle S^3 over the base manifold $\mathbb{CP}^1 \simeq S^2$ with the structure group $U(1) \simeq S^1$. They satisfy the following definition:

(i) The projection π called the *Hopf map* is defined by

$$\pi : S^3 \subset \mathbb{C}^2 \rightarrow \mathbb{CP}^1, \quad (z_1, z_2) \mapsto [z_1, z_2]. \quad (3.7)$$

(ii) The Lie group $U(1)$ acts on S^3 from the right as

$$S^3 \times U(1) \rightarrow S^3, \quad ((z_1, z_2), s) \mapsto (z_1, z_2)s = (z_1 s, z_2 s). \quad (3.8)$$

The $U(1)$ action on the fiber $\pi^{-1}([z_1, z_2]) \subset S^3$ is *simply transitive*, which means the following two properties:

(ii a) For any $(z_1, z_2) \in S^3$ and $s \in U(1)$, it satisfies $\pi((z_1, z_2)s) = \pi((z_1, z_2))$.

(ii b) If $\pi((z_1, z_2)) = \pi((z'_1, z'_2))$, then there exists a unique $s \in U(1)$ such that

$$(z'_1, z'_2) = (z_1, z_2)s. \quad (3.9)$$

(iii) The *section* of the Hopf fibration is defined by

$$\begin{aligned} \sigma_1 : U_1 &\rightarrow \pi^{-1}(U_1), & \sigma_1([z_1, z_2]) &= \frac{(z_1, z_2)|z_1|/z_1}{\sqrt{|z_1|^2 + |z_2|^2}}, \\ \sigma_2 : U_2 &\rightarrow \pi^{-1}(U_2), & \sigma_2([z_1, z_2]) &= \frac{(z_1, z_2)|z_2|/z_2}{\sqrt{|z_1|^2 + |z_2|^2}}. \end{aligned} \quad (3.10)$$

By the definition, the *transition function* can be read as

$$\psi_{12} : U_1 \cap U_2 \rightarrow U(1), \quad \psi_{12}([z_1, z_2]) = \frac{z_1}{|z_1|} \left(\frac{z_2}{|z_2|} \right)^{-1} \quad (3.11)$$

and it satisfies, on $U_1 \cap U_2$,

$$\psi_{12}([z_1, z_2])\psi_{21}([z_1, z_2]) = 1, \quad \sigma_2([z_1, z_2]) = \sigma_1([z_1, z_2])\psi_{12}([z_1, z_2]). \quad (3.12)$$

The total space S^3 of the Hopf fibration could be locally regarded as a direct product of the base manifold $\mathbb{CP}^1$ and the fiber $U(1)$ due to the *local trivialization* given by

$$\begin{aligned} \phi_1 : \pi^{-1}(U_1) &\rightarrow U_1 \times U(1), \quad \phi_1((z_1, z_2)) = ([z_1, z_2], \frac{z_1}{|z_1|}), \\ \phi_2 : \pi^{-1}(U_2) &\rightarrow U_2 \times U(1), \quad \phi_2((z_1, z_2)) = ([z_1, z_2], \frac{z_2}{|z_2|}). \end{aligned} \quad (3.13)$$

3.2 Canonical connection

Suppose that the hermitian inner product on $\mathbb{C}^2$ is defined by

$$\langle (z_1, z_2), (w_1, w_2) \rangle = \bar{z}_1 w_1 + \bar{z}_2 w_2 \quad \text{for } (z_1, z_2), (w_1, w_2) \in \mathbb{C}^2, \quad (3.14)$$

where $\bar{z}$ stands for the complex conjugate of $z \in \mathbb{C}$.

Definition 3.2. The *canonical connection* (or Berry-Simon connection) ω_i ($i = 1, 2$) is a $\mathfrak{u}(1) = i\mathbb{R}$ valued one-form on $U_i \subset \mathbb{CP}^1$ defined by

$$\begin{aligned} \omega_1([z_1, z_2]) &= \langle \sigma_1([z_1, z_2]), d\sigma_1([z_1, z_2]) \rangle, \\ \omega_2([z_1, z_2]) &= \langle \sigma_2([z_1, z_2]), d\sigma_2([z_1, z_2]) \rangle. \end{aligned} \quad (3.15)$$

They are explicitly given by

$$\begin{aligned} \omega_1([z_1, z_2]) &= \frac{|z_2|^2}{|z_1|^2 + |z_2|^2} \left[\frac{|z_2|}{z_2} d\left(\frac{z_2}{|z_2|}\right) - \frac{|z_1|}{z_1} d\left(\frac{z_1}{|z_1|}\right) \right], \\ \omega_2([z_1, z_2]) &= \frac{-|z_1|^2}{|z_1|^2 + |z_2|^2} \left[\frac{|z_2|}{z_2} d\left(\frac{z_2}{|z_2|}\right) - \frac{|z_1|}{z_1} d\left(\frac{z_1}{|z_1|}\right) \right]. \end{aligned} \quad (3.16)$$

Proposition 3.3. The canonical connections (Berry-Simon connections) ω_i ($i = 1, 2$) are mutually related by the gauge transformation with the transition function (3.11) on $U_1 \cap U_2$ as

$$\begin{aligned} \omega_2([z_1, z_2]) &= \omega_1([z_1, z_2]) + \psi_{12}([z_1, z_2])^{-1} d\psi_{12}([z_1, z_2]), \\ \omega_1([z_1, z_2]) &= \omega_2([z_1, z_2]) + \psi_{21}([z_1, z_2])^{-1} d\psi_{21}([z_1, z_2]). \end{aligned} \quad (3.17)$$

Proof. Since the second relation follows from the first one, it is sufficient to show the first relation. By the definition of the transition function (3.11), we have

$$\begin{aligned}\psi_{12}([z_1, z_2])^{-1} d\psi_{12}([z_1, z_2]) &= d \log \psi_{12}([z_1, z_2]) = \frac{|z_1|}{z_1} d \left(\frac{z_1}{|z_1|} \right) - \frac{|z_2|}{z_2} d \left(\frac{z_2}{|z_2|} \right) \\ &= \omega_2([z_1, z_2]) - \omega_1([z_1, z_2]).\end{aligned}\quad (3.18)$$

This is the desired relation. $\square$

In the above proof, we see that the following equality holds on $\pi^{-1}(U_1 \cap U_2)$

$$\omega_1([z_1, z_2]) + \frac{|z_1|}{z_1} d \left(\frac{z_1}{|z_1|} \right) = \omega_2([z_1, z_2]) + \frac{|z_2|}{z_2} d \left(\frac{z_2}{|z_2|} \right). \quad (3.19)$$

In other words, introducing $\mathfrak{u}(1)$ valued one-form on $\pi^{-1}(U_i) \simeq U_i \times U(1)$ by

$$\tilde{\omega}_i((z_1, z_2)) = \omega_i([z_1, z_2]) + \frac{|z_i|}{z_i} d \left(\frac{z_i}{|z_i|} \right), \quad (i = 1, 2) \quad (3.20)$$

we obtain

$$\tilde{\omega}_1((z_1, z_2)) = \tilde{\omega}_2((z_1, z_2)) \quad \text{on} \quad \pi^{-1}(U_1 \cap U_2). \quad (3.21)$$

Thus, the $\mathfrak{u}(1)$ valued one-form ω given as below is globally defined on $\mathbb{CP}^1$.

Definition-Proposition 3.4 (Canonical connection). *Define $\mathfrak{u}(1)$ valued one-form on $\pi^{-1}(U_i) \subset S^3$ ($i = 1, 2$) by*

$$\omega((z_1, z_2)) = \omega_i([z_1, z_2]) + \frac{|z_i|}{z_i} d \left(\frac{z_i}{|z_i|} \right). \quad (3.22)$$

Then, one-form ω is globally defined on S^3 . Explicitly, it is given by

$$\begin{aligned}\omega((z_1, z_2)) &= \langle (z_1, z_2), d(z_1, z_2) \rangle \\ &= \frac{1}{2} (\bar{z}_1 dz_1 + \bar{z}_2 dz_2 - z_1 d\bar{z}_1 - z_2 d\bar{z}_2) \quad \text{for} \quad (z_1, z_2) \in S^3.\end{aligned}\quad (3.23)$$

We refer to ω as the canonical connection (or Berry-Simon connection) on S^3 .

Note that the second equality in (3.23) follows by differentiating $\langle (z_1, z_2), (z_1, z_2) \rangle = 1$. Schematically, the connections ω and ω_i ($i = 1, 2$) are related as

$$\omega = \pi^*(\omega_i) \quad \text{and} \quad \sigma_i^*(\omega) = \omega_i. \quad (3.24)$$

3.3 Horizontal and vertical subspaces

The connection one-form ω defined in (3.23) satisfies two important properties, as stated below. For an element of Lie algebra $A \in \mathfrak{u}(1)$, consider the one-parameter subgroup

$$\{ e^{tA} \mid t \in \mathbb{R} \} \subset S^3. \quad (3.25)$$

Let us make this subgroup act on $(z_1, z_2) \in S^3$ from right and define the vector $(A^*)_{(z_1, z_2)} \in T_{(z_1, z_2)}S^3$ as the tangent vector of the orbit $(z_1, z_2)e^{tA}$ at $t = 0$,

$$(A^*)_{(z_1, z_2)} = \frac{d}{dt} ((z_1, z_2)e^{tA}) \Big|_{t=0} = (z_1, z_2)A. \quad (3.26)$$

For $A = i\varphi \in \mathfrak{u}(1)$ with $\varphi \in \mathbb{R}$, it is explicitly read as

$$(A^*)_{(z_1, z_2)} = (z_1, z_2)i\varphi = i\varphi (z_1\partial_{z_1} + z_2\partial_{z_2} - \bar{z}_1\partial_{\bar{z}_1} - \bar{z}_2\partial_{\bar{z}_2}) (z_1, z_2). \quad (3.27)$$

Here, the complex conjugate terms are needed for the reality condition for A^* .

Definition 3.5 (Fundamental vector field). The vector field $A^* \in \mathfrak{X}(S^3)$ is said to be the *fundamental vector field* associated with $A = i\varphi \in \mathfrak{u}(1)$. It is explicitly defined by

$$(A^*)_{(z_1, z_2)} = i\varphi (z_1\partial_{z_1} + z_2\partial_{z_2} - \bar{z}_1\partial_{\bar{z}_1} - \bar{z}_2\partial_{\bar{z}_2}) \quad \text{at} \quad (z_1, z_2) \in S^3. \quad (3.28)$$

We denote the map by $v : \mathfrak{u}(1) \rightarrow \mathfrak{X}(S^3)$, $A \mapsto v(A) = A^*$.

For $a \in U(1)$, let us denote the right action on $S^3 \subset \mathbb{C}^2$ by

$$R_a : S^3 \rightarrow S^3, \quad (z_1, z_2) \mapsto R_a(z_1, z_2) = (z_1, z_2)a. \quad (3.29)$$

Proposition 3.6. *The canonical connection ω on S^3 in (3.23) satisfies the following conditions:*

$$(a) \quad \omega(A^*) = A \quad \text{for any} \quad A \in \mathfrak{u}(1),$$

$$(b) \quad (R_a)^*\omega = \omega \quad \text{for any} \quad a \in U(1).$$

Proof. (a): By the definitions of ω in (3.23) and A^* for $A = i\varphi$ ($\varphi \in \mathbb{R}$) in (3.28), it is directly calculated as

$$\begin{aligned} \omega(A^*) &= \langle (z_1, z_2), A^*(z_1, z_2) \rangle = \langle (z_1, z_2), i\varphi(z_1, z_2) \rangle \\ &= (z_1\bar{z}_1 + z_2\bar{z}_2) i\varphi. \end{aligned} \quad (3.30)$$

Because of $|z_1|^2 + |z_2|^2 = 1$, we obtain $\omega(A^*) = i\varphi = A$.

(b): By the definition of ω in (3.23), we see that

$$\begin{aligned} (R_a)^*\omega((z_1, z_2)) &= \omega((z_1, z_2)a) = \langle (z_1, z_2)a, d(z_1, z_2)a \rangle \\ &= \bar{a}\langle (z_1, z_2), d(z_1, z_2) \rangle a = \bar{a}\omega((z_1, z_2))a. \end{aligned} \quad (3.31)$$

Since $\bar{a}a = |a|^2 = 1$ for $a \in U(1)$, we have $(R_a)^*\omega = \omega$. $\square$

The following proposition means that the connection one-form ω is a projector of TS^3 onto the subspace, so-called the horizontal subspace.

Definition-Proposition 3.7 (Horizontal and vertical subspaces). *Define the subspaces $H_{(z_1, z_2)}$ and $V_{(z_1, z_2)}$ of $T_{(z_1, z_2)}S^3$ by*

$$\begin{aligned} H_{(z_1, z_2)} &= \text{Ker } \omega = \{ X \in T_{(z_1, z_2)}S^3 \mid \omega(X) = 0 \}, \\ V_{(z_1, z_2)} &= v(\mathfrak{u}(1)) = \{ (A^*)_{(z_1, z_2)} \in T_{(z_1, z_2)}S^3 \mid A \in \mathfrak{u}(1) \}. \end{aligned} \quad (3.32)$$

These are referred to as the horizontal and vertical subspaces, respectively. Then, it holds that

- (a) $T_{(z_1, z_2)}S^3 = V_{(z_1, z_2)} \oplus H_{(z_1, z_2)}$ (direct sum),
- (b) $(R_a)_*H_{(z_1, z_2)} = H_{(z_1, z_2)a}$ for any $a \in U(1)$.

Proof. (a): By the definition of $v : \mathfrak{u}(1) \rightarrow \mathfrak{X}(S^3)$, $A \mapsto A^*$ in (3.28) and the property of ω in Prop. 3.6 (a), the composition map

$$\omega \circ v : \mathfrak{u}(1) \rightarrow T_{(z_1, z_2)}S^3 \rightarrow \mathfrak{u}(1), \quad (\omega \circ v)(A) = \omega((A^*)_{(z_1, z_2)}) = A \quad (3.33)$$

is the identity map on $\mathfrak{u}(1)$. Hence, the map v is injective, and ω is surjective. By the definition of $H_{(z_1, z_2)} = \text{Ker } \omega$, we have the following short exact sequence:

$$0 \longrightarrow H_{(z_1, z_2)} \longrightarrow T_{(z_1, z_2)}S^3 \xrightarrow{\omega} \mathfrak{u}(1) \longrightarrow 0. \quad (3.34)$$

By the injectivity of $v : \mathfrak{u}(1) \rightarrow T_{(z_1, z_2)}S^3$, this sequence splits and we have

$$T_{(z_1, z_2)}S^3 = v(\mathfrak{u}(1)) \oplus H_{(z_1, z_2)} = V_{(z_1, z_2)} \oplus H_{(z_1, z_2)}. \quad (3.35)$$

(b): For $a \in U(1)$, an arbitrary $Y \in (R_a)_*H_{(z_1, z_2)}$ could be written as $Y = (R_a)_*X = Xa$ with $X \in H_{(z_1, z_2)}$. By Prop. 3.6 (b), we have

$$\omega(Y) = \omega((R_a)_*X) = ((R_a)^*\omega)(X) = \omega(X) = 0, \quad (3.36)$$

which means that $Y \in H_{(z_1, z_2)a}$ and, hence, $(R_a)_*H_{(z_1, z_2)} \subset H_{(z_1, z_2)a}$. Conversely, for any $Xa \in H_{(z_1, z_2)a}$ with $X \in T_{(z_1, z_2)}S^3$, it holds that

$$0 = \omega(Xa) = \omega((R_a)_*X) = ((R_a)^*\omega)(X) = \omega(X). \quad (3.37)$$

This implies that $X \in H_{(z_1, z_2)}$ and $(R_a)_*H_{(z_1, z_2)} \supset H_{(z_1, z_2)a}$. This proves that

$$(R_a)_*H_{(z_1, z_2)} = H_{(z_1, z_2)a}. \quad (3.38)$$

□

Remark 3.8. Def.-Prop. 3.7 together with Prop. 3.6 are often adopted as the definition of the connection one-form for the generic principal fiber bundles with the structure group, a Lie group G , where Prop. 3.6 (b) should be replaced by $(R_a)^*\omega = a^{-1}\omega a$ with $a \in G$. For details, see [9, 10].

At the end of this section, we present concrete descriptions of the horizontal and vertical subspaces in terms of local coordinates. Set

$$z_1 = e^{i\varphi_1} \cos \frac{\beta}{2}, \quad z_2 = e^{i\varphi_2} \sin \frac{\beta}{2}, \quad (3.39)$$

where $\varphi_1, \varphi_2, \beta \in \mathbb{R}$ with $0 \leq \varphi_1, \varphi_2 < 2\pi$ and $0 \leq \beta \leq \pi$. Then, $(z_1, z_2) \in S^3$ because of

$$|z_1|^2 + |z_2|^2 = 1. \quad (3.40)$$

The canonical connection ω on S^3 is calculated as

$$\begin{aligned} \omega &= \bar{z}_1 dz_1 + \bar{z}_2 dz_2 = i \cos^2 \frac{\beta}{2} d\varphi_1 + i \sin^2 \frac{\beta}{2} d\varphi_2 \\ &= \frac{i}{2} (d\varphi + \cos \beta d\theta). \end{aligned} \quad (3.41)$$

In the last equality, we have set

$$\varphi = \varphi_1 + \varphi_2, \quad \theta = \varphi_1 - \varphi_2. \quad (3.42)$$

The horizontal and vertical subspaces given in Def.-Prop. 3.7 are explicitly described as follows.

Corollary 3.9. *For the canonical connection ω on S^3 ,*

$$\omega((z_1, z_2)) = \frac{1}{2} (\bar{z}_1 dz_1 + \bar{z}_2 dz_2 - z_1 d\bar{z}_1 - z_2 d\bar{z}_2) \quad \text{with} \quad (z_1, z_2) \in S^3, \quad (3.43)$$

the tangent space $T_{(z_1, z_2)} S^3$ is decomposed into the direct sum of two-dimensional horizontal subspace $H_{(z_1, z_2)}$ and one-dimensional vertical subspace $V_{(z_1, z_2)}$. Their bases are given by

$$\begin{aligned} H_{(z_1, z_2)} &= \langle i|z_2|^2(z_1\partial_{z_1} - \bar{z}_1\partial_{\bar{z}_1}) - i|z_1|^2(z_2\partial_{z_2} - \bar{z}_2\partial_{\bar{z}_2}), \\ &\quad - \frac{|z_2|}{|z_1|}(z_1\partial_{z_1} + \bar{z}_1\partial_{\bar{z}_1}) + \frac{|z_1|}{|z_2|}(z_2\partial_{z_2} + \bar{z}_2\partial_{\bar{z}_2}) \rangle, \\ V_{(z_1, z_2)} &= \langle i(z_1\partial_{z_1} + z_2\partial_{z_2} - \bar{z}_1\partial_{\bar{z}_1} - \bar{z}_2\partial_{\bar{z}_2}) \rangle. \end{aligned} \quad (3.44)$$

In terms of the spherical coordinates,

$$z_1 = e^{\frac{i}{2}(\varphi+\theta)} \cos \frac{\beta}{2}, \quad z_2 = e^{\frac{i}{2}(\varphi-\theta)} \sin \frac{\beta}{2}, \quad (3.45)$$

the canonical connection and the horizontal and vertical subspaces are simply described as

$$\omega((z_1, z_2)) = \frac{i}{2} (d\varphi + \cos \beta d\theta) , \quad (3.46)$$

$$\begin{aligned} H_{(z_1, z_2)} &= \langle \partial_\theta - \cos \beta \partial_\varphi, \partial_\beta \rangle , \\ V_{(z_1, z_2)} &= \langle \partial_\varphi \rangle . \end{aligned} \quad (3.47)$$

Proof. The expression of the canonical one-form (3.46) follows from the definition (3.43) by the coordinate change (3.45). We first show (3.47). An arbitrary vector $X \in T_{(z_1, z_2)} S^3$ could be written as

$$X = f\partial_\theta + g\partial_\beta + h\partial_\varphi \quad (3.48)$$

with some real functions f, g, h on S^3 . If $X \in H_{(z_1, z_2)} = \text{Ker } \omega$, then $\omega(X) = 0$ implies that $\cos \beta f + h = 0$. Hence, any $X \in H_{(z_1, z_2)}$ is written as

$$X = f(\partial_\theta - \cos \beta \partial_\varphi) + g\partial_\beta . \quad (3.49)$$

Since two vectors $\partial_\theta - \cos \beta \partial_\varphi$ and ∂_β are linearly independent, they form the basis of $H_{(z_1, z_2)}$. On the other hand, the vertical subspace $V_{(z_1, z_2)} = v(\mathfrak{u}(1))$ is spanned by the fundamental vector field A^* associated with $A = ia \in \mathfrak{u}(1)$ with some $a \in \mathbb{R}$. By the definition of A^* in (3.28) and the spherical coordinates (3.45), it is calculated as

$$A^* = ia(z_1\partial_{z_1} + z_2\partial_{z_2} - \bar{z}_1\partial_{\bar{z}_1} - \bar{z}_2\partial_{\bar{z}_2}) = 2a\partial_\varphi . \quad (3.50)$$

Hence, $V_{(z_1, z_2)}$ is generated by ∂_φ .

Next, the expressions (3.44) are deduced from (3.47) via the coordinate change (3.45). More concretely, substituting the differential operators

$$\begin{aligned} \partial_\varphi &= \frac{i}{2} (z_1\partial_{z_1} + z_2\partial_{z_2} - \bar{z}_1\partial_{\bar{z}_1} - \bar{z}_2\partial_{\bar{z}_2}) , \\ \partial_\theta &= \frac{i}{2} (z_1\partial_{z_1} - z_2\partial_{z_2} - \bar{z}_1\partial_{\bar{z}_1} + \bar{z}_2\partial_{\bar{z}_2}) , \\ \partial_\beta &= -\frac{|z_2|}{2|z_1|} (z_1\partial_{z_1} + \bar{z}_1\partial_{\bar{z}_1}) + \frac{|z_1|}{2|z_2|} (z_2\partial_{z_2} + \bar{z}_2\partial_{\bar{z}_2}) , \end{aligned} \quad (3.51)$$

for (3.47), we obtain (3.44). This completes the proof. $\square$

3.4 Horizontal lift and parallel displacement

Let us now introduce the notion of the *horizontal lift* of a curve in $\mathbb{CP}^1$ to S^3 and the *parallel displacement* along the horizontal curve. The latter is also commonly referred to as *parallel transport* in the physics and mathematical physics literature. We shall introduce them for the Hopf fiber bundle endowed with the canonical connection.

Definition 3.10 (Horizontal lift). Let c be a piecewise differentiable curve of class C^1 in $\mathbb{CP}^1$ described as $c = \{ x(t) \in \mathbb{CP}^1 \mid t \in [a, b] \} \subset \mathbb{CP}^1$. A *horizontal lift* of c is a curve $\tilde{c}$ in S^3 given by

$$\tilde{c} = \{ \tilde{x}(t) \in S^3 \mid t \in [a, b] \} \subset S^3, \quad (3.52)$$

which satisfies the following two conditions for any $t \in [a, b]$;

- (i) $\pi(\tilde{x}(t)) = x(t)$,
- (ii) $\frac{d}{dt}\tilde{x}(t) \in H_{\tilde{x}(t)}$.

If the curve $\tilde{c} \subset S^3$ satisfies either (i) or (ii), it is said to be a *lift* of c or *horizontal*, respectively.

For the Hopf fiber bundle with the canonical connection $\omega = \langle (z_1, z_2), d(z_1, z_2) \rangle$, the condition of horizontal lift of a curve is explicitly described as follows.

Proposition 3.11. *Let c be a piecewise differentiable curve of class C^1 in $\mathbb{CP}^1$ given by*

$$c = \{ [z_1(t), z_2(t)] \in \mathbb{CP}^1 \mid t \in [a, b] \} \subset \mathbb{CP}^1. \quad (3.53)$$

Then, a curve

$$\tilde{c} = \{ (z_1(t), z_2(t)) \in S^3 \mid t \in [a, b] \} \subset S^3, \quad (3.54)$$

is a horizontal lift of a curve c if it satisfies that $\pi((z_1(t), z_2(t))) = [z_1(t), z_2(t)]$ and

$$\bar{z}_1 \dot{z}_1 + \bar{z}_2 \dot{z}_2 = 0. \quad (3.55)$$

Here, the dot means the derivative with respect to t , i.e., $\dot{x} = dx/dt$.

Remark 3.12. In terms of the parametrization of S^3 in (3.45), the horizontal condition can be written as

$$\bar{z}_1 \dot{z}_1 + \bar{z}_2 \dot{z}_2 = \frac{i}{2}(\dot{\varphi} + \cos \beta \dot{\theta}) = 0. \quad (3.56)$$

The angular variable φ in this expression is the Hopf coordinate. It should not be confused with a local fiber coordinate associated with a choice of local section of the Hopf bundle. Precisely, over the local charts $U_1, U_2 \subset \mathbb{CP}^1$ one may write

$$\varphi + \theta = 2\chi_1, \quad \varphi - \theta = 2\chi_2, \quad (3.57)$$

where χ_1, χ_2 are the fiber coordinates relative to the local sections, respectively.

Proof. To show a curve $\tilde{c} \subset S^3$ is a horizontal lift of $c \subset \mathbb{CP}^1$, it is sufficient to prove the horizontal condition (ii) of Def. 3.10,

$$\frac{d}{dt}(z_1(t), z_2(t)) \in H_{(z_1(t), z_2(t))}, \quad (3.58)$$

as it is a lift of c , $\pi((z_1(t), z_2(t))) = [z_1(t), z_2(t)]$. Since we have

$$\omega((\dot{z}_1, \dot{z}_2)) = \langle (z_1, z_2), (\dot{z}_1, \dot{z}_2) \rangle = \bar{z}_1 \dot{z}_1 + \bar{z}_2 \dot{z}_2, \quad (3.59)$$

the condition $\omega((\dot{z}_1, \dot{z}_2)) = 0$ is equivalent to that

$$\bar{z}_1 \dot{z}_1 + \bar{z}_2 \dot{z}_2 = 0. \quad (3.60)$$

The second equality follows from the coordinate change (3.45). This completes the proof. $\square$

Definition-Proposition 3.13 (Parallel displacement). *For a fixed $x_0 \in \mathbb{CP}^1$, take $y_0 \in \pi^{-1}(x_0) \subset S^3$. Then, there uniquely exists a horizontal curve $\tilde{c}$ in S^3 starting from $y_0 \in S^3$, that is,*

$$\tilde{c} = \{ \tilde{x}(t) \in S^3 \mid \omega(\dot{\tilde{x}}) = 0, \tilde{x}(a) = y_0, t \in [a, b] \} \subset S^3. \quad (3.61)$$

The curve $\tilde{c}$ is said to be the parallel displacement of y_0 along the curve c .

Proof. By Prop. 3.11, the existence and uniqueness of a curve $\tilde{c}$ reduces to those of a solution of the first-order ordinary differential equation of φ in (3.55). Since the differential equation (3.55) with the initial condition $\tilde{x}(a) = y_0$ has a unique solution, the claim follows. $\square$

4 Rotation angle as the holonomy

The notion of the Hopf fibration equipped with the canonical connection (3.23) allows us to describe the geometric phase Δ_g in (2.14) in terms of the holonomy of the Hopf fibration (Thm. 4.5 in § 4.1). To state the relation precisely, we need to take into account the representation (Def. 4.4),

$$\rho_2 : U(1) \longrightarrow U(1), \quad \rho_2(e^{i\varphi}) = e^{2i\varphi}.$$

The canonical Hopf connection gives the $U(1)$ holonomy, whereas the geometric phase of the rotating disc is obtained from this holonomy through the natural weight-two representation

of the fiber group. Conceptually, ρ_2 may be viewed as a simple *functorial bridge* from the mathematical holonomy of the Hopf fibration to the physical rotation angle observed in the model. Therefore, if $\tau(\gamma) \in U(1)$ denotes the holonomy along the Gauss curve γ , the correct relation is

$$\rho_2(\tau(\gamma)) = \exp(i\Delta_g).$$

In this sense, the geometric phase Δ_g is described as the holonomy of the Hopf fibration, after applying the representation ρ_2 . Furthermore, we will explain in § 4.2 that the information carried by the topological indices is retained at the level of the real lift $\mathbb{R} \rightarrow U(1)$, although it is lost after exponentiation to $U(1)$.

4.1 $U(1)$ holonomy and the geometric phase

$U(1)$ holonomy

Suppose that $c = \{ x(t) \in \mathbb{CP}^1 \mid t \in [0, 1] \} \subset \mathbb{CP}^1$ is a closed curve such as $x(0) = x(1) = x_0$. For some $y_0 \in \pi^{-1}(x_0)$, let $\tilde{c}$ be the parallel displacement of y_0 along c described by

$$\tilde{c} = \{ \tilde{x}(t) \in S^3 \mid t \in [0, 1] \} \subset S^3, \quad (4.1)$$

where $\tilde{x}(0) = y_0$ and $\pi(y_0) = \pi(\tilde{x}(1)) = x_0$. Since both $\tilde{x}(0)$ and $\tilde{x}(1)$ belong to the same fiber $\pi^{-1}(x_0)$, there uniquely exists an element $\tau(c) \in U(1)$ such that

$$\tilde{x}(1) = \tilde{x}(0) \cdot \tau(c), \quad (4.2)$$

which is called the *holonomy* of c with a base point x_0 .

Since $U(1)$ action commutes with the parallel displacement, the holonomy is compatible with the concatenation of based loops:

$$\tau(c' \circ c) = \tau(c') \cdot \tau(c), \quad (4.3)$$

where c' is another closed curve based at x_0 defined by $c' = \{ u(t) \in \mathbb{CP}^1 \mid t \in [0, 1] \}$, and their composition is $c' \circ c = \{ v(t) \in \mathbb{CP}^1 \mid t \in [0, 1] \}$ by

$$v(t) = \begin{cases} x(2t) & (0 \leq t \leq \frac{1}{2}) \\ u(2t - 1) & (\frac{1}{2} \leq t \leq 1) \end{cases}. \quad (4.4)$$

In particular, we have $\tau(c^{-1} \circ c) = \tau(c \circ c^{-1}) = 1 \in U(1)$ where $c^{-1} = \{ x(1-t) \in \mathbb{CP}^1 \mid t \in [0, 1] \}$.

In general, the set $\{\tau(c)\}$ forms a subgroup of the structure group G , which is referred to as the *holonomy group* with base point x_0 . In the case of the Hopf fibration with $U(1)$ structure group, the holonomy group is isomorphic to $U(1)$ itself.

In the subsequent arguments, through the diffeomorphism of S^2 and $\mathbb{CP}^1$, which is summarized in App. A, we can regard the Gauss curve γ in (2.7) as a curve in $\mathbb{CP}^1$:

$$\gamma = \{ [z_1(t), z_2(t)] \in \mathbb{CP}^1 \mid t \in [0, 1] \} \subset \mathbb{CP}^1, \quad (4.5)$$

where $z_1(t) = e^{\frac{i}{2}\theta(t)} \cos \frac{\beta(t)}{2}$, $z_2(t) = e^{-\frac{i}{2}\theta(t)} \sin \frac{\beta(t)}{2}$.

Proposition 4.1. *Let γ be a piecewise smooth closed Gauss curve on $S^2 \simeq \mathbb{CP}^1$ associated with the motion (M), and let $\tau(\gamma) \in U(1)$ be the holonomy of the Hopf fibration $S^3 \rightarrow S^2$ equipped with the canonical connection. Then, the holonomy is given by*

$$\tau(\gamma) = (-1)^n \exp \left(-\frac{i}{2} \int_0^1 \cos \beta(t) \dot{\theta}(t) dt \right), \quad (4.6)$$

where $n = \theta(1)/2\pi \in \mathbb{Z}$ is the topological number defined in (T).

Remark 4.2. Although the two factors

$$(-1)^n \quad \text{and} \quad \exp \left(-\frac{i}{2} \int_0^1 \cos \beta(t) \dot{\theta}(t) dt \right) \quad (4.7)$$

on the right-hand side of (4.6) depend separately on the motion (M), their product depends only on the associated Gauss curve γ . Thus, their combination is an intrinsic quantity associated with γ , as it must be since it gives the holonomy $\tau(\gamma)$.

Proof of Proposition 4.1. Through the diffeomorphism of S^2 and $\mathbb{CP}^1$, which is summarized in App. A, we can regard the Gauss curve γ in (2.7) as a curve in $\mathbb{CP}^1$:

$$\gamma = \{ [z_1(t), z_2(t)] \in \mathbb{CP}^1 \mid t \in [0, 1] \} \subset \mathbb{CP}^1. \quad (4.8)$$

We first consider the case where the closed curve γ in (4.8) is contained in a single local chart. On the chart U_1 , where $z_1 \neq 0$ ($\beta \neq \pi$), a lift $\tilde{\gamma}$ can be parametrized as

$$\tilde{\gamma} = \{ \sigma_1(t) e^{i\chi_1(t)} \in S^3 \mid t \in [0, 1] \} \subset S^3$$

with $\sigma_1(t) = \left(\cos \frac{\beta(t)}{2}, e^{-i\theta(t)} \sin \frac{\beta(t)}{2} \right), \quad (4.9)$

and it is horizontal if it satisfies the condition

$$\omega\left(\frac{d}{dt}\right) = (\omega_1 + id\chi_1)\left(\frac{d}{dt}\right) = 0 \quad \text{with} \quad \omega_1 = \frac{i}{2}(-1 + \cos \beta) d\theta. \quad (4.10)$$

Hence, the holonomy is computed as

$$\begin{aligned}\tau(\gamma) &= e^{i(\chi_1(1)-\chi_1(0))} = \exp\left(i \int_{\gamma} d\chi_1\right) = \exp\left(- \int_{\gamma} \omega_1\right) = \exp\left(\frac{i}{2} \int_{\gamma} (1 - \cos \beta) d\theta\right) \\ &= \exp\left(\frac{i}{2} \int_0^1 (1 - \cos \beta(t)) \dot{\theta}(t) dt\right)\end{aligned}\quad (4.11)$$

Since the integration of θ gives the topological number (T),

$$\int_0^1 \dot{\theta}(t) dt = \theta(1) - \theta(0) = 2\pi n, \quad (4.12)$$

the holonomy reduces to the desired expression,

$$\tau(\gamma) = (-1)^n \exp\left(-\frac{i}{2} \int_0^1 \cos \beta(t) \dot{\theta}(t) dt\right). \quad (4.13)$$

Similarly, if γ is included in the chart U_2 , where $z_2 \neq 0$ ($\beta \neq 0$), a lift $\tilde{\gamma}$ can be parametrized as

$$\begin{aligned}\tilde{\gamma} &= \{ \sigma_2(t) e^{i\chi_2(t)} \in S^3 \mid t \in [0, 1] \} \subset S^3 \\ \text{with } \sigma_2(t) &= \left(e^{i\theta(t)} \cos \frac{\beta(t)}{2}, \sin \frac{\beta(t)}{2} \right),\end{aligned}\quad (4.14)$$

and it is horizontal if it satisfies the condition

$$\omega\left(\frac{d}{dt}\right) = (\omega_2 + i d\chi_2)\left(\frac{d}{dt}\right) = 0 \quad \text{with} \quad \omega_2 = \frac{i}{2}(1 + \cos \beta) d\theta. \quad (4.15)$$

In this case, the holonomy is computed as

$$\begin{aligned}\tau(\gamma) &= e^{i(\chi_2(1)-\chi_2(0))} = \exp\left(i \int_{\gamma} d\chi_2\right) = \exp\left(- \int_{\gamma} \omega_2\right) = \exp\left(-\frac{i}{2} \int_{\gamma} (1 + \cos \beta) d\theta\right) \\ &= \exp\left(-\frac{i}{2} \int_0^1 (1 + \cos \beta(t)) \dot{\theta}(t) dt\right)\end{aligned}\quad (4.16)$$

Evaluating the integration of θ , we also have the same expression (4.6).

Now consider the general case where γ is *not* contained in a single local chart. Let us decompose the time interval as

$$0 = t_0 < t_1 < \cdots < t_N = 1, \quad (4.17)$$

such that each associated segment

$$\gamma_j = \{ [z_1(t), z_2(t)] \in \mathbb{CP}^1 \mid t \in [t_{j-1}, t_j] \} \quad \text{with} \quad j = 1, \dots, N, \quad (4.18)$$

is contained in one chart U_{α_j} , where $\alpha_j \in \{1, 2\}$. Choose the subdivision so that the initial and final segments are contained in the same chart containing the base point at $t = 0$ and $t = 1$. By combining adjacent segments contained in the same chart, we may assume that the charts alternate in the following ordered decomposition of γ ,

$$\gamma = \gamma_1 \cup \gamma_2 \cup \cdots \cup \gamma_N. \quad (4.19)$$

More precisely, we suppose that N is odd and that $\gamma_j \subset U_1$ for odd j , while $\gamma_j \subset U_2$ for even j . Thus, both γ_1 and γ_N are contained in U_1 . Note that the subsequent argument is parallel when they are in U_2 .

Let us denote the horizontal lift of γ_j by

$$\tilde{\gamma}_j = \{ \sigma_{\alpha_j}(t) e^{i\chi_{\alpha_j}(t)} \in S^3 \mid t \in [t_{j-1}, t_j] \} \quad \text{with} \quad j = 1, \dots, N. \quad (4.20)$$

The holonomy along γ is then given by an ordered product in which the local parallel transport operators along the segments and the transition functions at the switching points appear alternately. Since it holds that

$$\sigma_2(t) = \sigma_1(t) \psi_{12}(t), \quad \psi_{12}(t) = \psi_{21}(t)^{-1} = e^{i\theta(t)} \quad (4.21)$$

at the connection points, we obtain

$$\begin{aligned} \tau(\gamma) &= \exp \left(i \int_{\gamma_1} d\chi_1 \right) \psi_{21}(t_1) \exp \left(i \int_{\gamma_2} d\chi_2 \right) \psi_{12}(t_2) \cdots \psi_{12}(t_{N-1}) \exp \left(i \int_{\gamma_N} d\chi_1 \right) \\ &= \exp \left(- \int_{\gamma_1} \omega_1 \right) e^{-i\theta(t_1)} \exp \left(- \int_{\gamma_2} \omega_2 \right) e^{i\theta(t_2)} \cdots e^{i\theta(t_{N-1})} \exp \left(- \int_{\gamma_N} \omega_1 \right), \end{aligned} \quad (4.22)$$

where the second equality follows from the horizontal conditions. Using the explicit expressions ω_1 and ω_2 , we have

$$\begin{aligned} \tau(\gamma) &= e^{\frac{i}{2}(\theta(t_1) - \theta(t_0))} e^{-i\theta(t_1)} e^{-\frac{i}{2}(\theta(t_2) - \theta(t_1))} e^{i\theta(t_2)} \cdots e^{i\theta(t_{N-1})} e^{\frac{i}{2}(\theta(t_N) - \theta(t_{N-1}))} \\ &\quad \times \exp \left(- \frac{i}{2} \sum_{j=1}^N \int_{t_{j-1}}^{t_j} \cos \beta(t) \dot{\theta}(t) dt \right) \\ &= e^{\frac{i}{2}(\theta(t_N) - \theta(t_0))} \exp \left(- \frac{i}{2} \int_0^1 \cos \beta(t) \dot{\theta}(t) dt \right). \end{aligned} \quad (4.23)$$

By the topological condition (T),

$$\theta(t_N) - \theta(t_0) = \theta(1) - \theta(0) = 2\pi n. \quad (4.24)$$

Hence, it reduces to

$$\tau(\gamma) = (-1)^n \exp \left(- \frac{i}{2} \int_0^1 \cos \beta(t) \dot{\theta}(t) dt \right). \quad (4.25)$$

Thus, this completes the proof. $\square$

Remark 4.3. If γ is simple, then, in terms of the oriented spherical areas defined in (2.13), Prop. 4.1 may equivalently be expressed as

$$\tau(\gamma) = \exp\left(\frac{i}{2}A_+\right) = \exp\left(-\frac{i}{2}A_-\right). \quad (4.26)$$

Thus, the holonomy has a direct geometric interpretation as the exponential of one half of the oriented spherical area.

From $U(1)$ holonomy to the geometric phase

The geometric phase is not identified directly with the $U(1)$ holonomy itself, but through a representation of the fiber group. To state the relation precisely, let us define the *functorial bridge from mathematics* of the Hopf fibration *to physics* of the rotation angle.

Definition 4.4. Define the weight-two representation of the $U(1)$ fiber by

$$\rho_2 : U(1) \longrightarrow U(1), \quad \rho_2(u) = u^2. \quad (4.27)$$

This weight-two representation allows us to connect the geometric phase of our model with the holonomy of the Hopf fibration.

Theorem 4.5. Let γ be a piecewise smooth Gauss curve on $\mathbb{CP}^1$ associated with the motion (M). The geometric phase Δ_g and the holonomy $\tau(\gamma)$ are related as

$$\rho_2(\tau(\gamma)) = \exp(i\Delta_g). \quad (4.28)$$

Proof. By Prop. 4.1 and the geometric phase (2.4), it holds that

$$\tau(\gamma) = (-1)^n \exp\left(\frac{i}{2}\Delta_g\right). \quad (4.29)$$

Therefore, applying the weight-two representation (4.27), we obtain

$$\rho_2(\tau(\gamma)) = \tau(\gamma)^2 = \exp(i\Delta_g), \quad (4.30)$$

since $(-1)^{2n} = 1$. □

4.2 Geometric phase and a real lift of the $U(1)$ holonomy

For a simple Gauss curve γ , the careful reader might notice that, in the description of holonomy, the contributions of topological indices $I_\pm \in \{0, 1, 2\}$ for $\Delta_g = A_+ - 2\pi I_+$ are hidden since

$$\rho_2(\tau(\gamma)) = \exp(i\Delta_g) = e^{i(A_+ - 2\pi I_+)} = e^{iA_+}. \quad (4.31)$$

In this sense, it is useful to consider a real lift of the $U(1)$ holonomy in order to retain the real-valued information contained in the geometric phase.

In other words, Δ_g may be regarded as an element of the universal covering space $\mathbb{R}$ over $S^1 \simeq U(1)$, that is

$$\begin{aligned} 0 \longrightarrow 2\pi\mathbb{Z} \longrightarrow \mathbb{R} \xrightarrow{p} U(1) \longrightarrow 1, \\ x \longmapsto \exp(ix). \end{aligned} \quad (4.32)$$

This viewpoint is also natural from the Lie-algebraic perspective³: the connection one-form takes values in the Lie algebra $\mathfrak{u}(1) = i\mathbb{R}$, and the passage to the $U(1)$ holonomy is made through exponentiation, which identifies real phases differing by 2π .

To state the claim precisely, we introduce the lift of the representation ρ_2 in Def. 4.4 to the universal covering space of $U(1)$, which will be used to relate a lifted Hopf holonomy to the real-valued geometric phase.

Definition 4.6. Let

$$p : \mathbb{R} \rightarrow U(1), \quad p(x) = e^{ix} \quad (4.33)$$

be the universal covering map. Define a natural lift $\tilde{\rho}_2$ of ρ_2 in Def. 4.4 by

$$\tilde{\rho}_2 : \mathbb{R} \rightarrow \mathbb{R}, \quad \tilde{\rho}_2(x) = 2x, \quad (4.34)$$

satisfying $p \circ \tilde{\rho}_2 = \rho_2 \circ p$, which means that these maps fit into the commutative diagram,

$$\begin{array}{ccc} \mathbb{R} & \xrightarrow{\tilde{\rho}_2} & \mathbb{R} \\ p \downarrow & & \downarrow p \\ U(1) & \xrightarrow{\rho_2} & U(1). \end{array}$$

Let us introduce a *lifted holonomy* $\tilde{\tau}_M(\gamma) \in \mathbb{R}$ satisfying the property

$$p(\tilde{\tau}_M(\gamma)) = e^{i\tilde{\tau}_M(\gamma)} = \tau(\gamma), \quad (4.35)$$

where p is the covering map. The subscript M emphasizes that this real lift depends on the motion (M), or equivalently on the choice of regularization, and not only on the limiting curve γ . This condition alone, however, determines $\tilde{\tau}_M(\gamma)$ only up to an additive multiple of 2π . In the present setting, we choose the following representative, which is naturally adapted to the expression obtained in the proof of Prop. 4.1, that is,

$$\tilde{\tau}_M(\gamma) = \pi n - \frac{1}{2} \int_0^1 \cos \beta(t) \dot{\theta}(t) dt. \quad (4.36)$$

³The author thanks Prof. Yasukura for pointing this out.

Then, it is easy to see that it matches the condition (4.35). The choice of $\tilde{\tau}_M(\gamma)$ is understood modulo $2\pi\mathbb{Z}$. In particular, another choice of local trivialization may give a representative differing from the above by an integral multiple of 2π .

By the expression of the geometric phase (2.4), it is equal to

$$\tilde{\tau}_M(\gamma) = \pi n + \frac{1}{2}\Delta_g. \quad (4.37)$$

Therefore, using the lifted weight-two representation (4.34), we obtain

$$\tilde{\rho}_2(\tilde{\tau}_M(\gamma)) = 2\tilde{\tau}_M(\gamma) = 2\pi n + \Delta_g. \quad (4.38)$$

Thus the geometric phase Δ_g is not merely the image of the Hopf holonomy in $U(1)$, but the image of its lifted holonomy under the lifted representation $\tilde{\rho}_2$. In this lifted formulation, the topological indices $I_{\pm}$ are retained:

$$\Delta_g = A_+ - 2\pi I_+ = -A_- + 2\pi I_-. \quad (4.39)$$

After exponentiation of (4.38), the topological terms disappear, and the result in Thm. 4.5 is restored,

$$\rho_2(\tau(\gamma)) = \exp(i\Delta_g). \quad (4.40)$$

Remark 4.7. The weight two appearing in ρ_2 is not an artifact of the normalization of the angular coordinates. Its geometric origin is the double covering $SU(2) \rightarrow SO(3)$. The Hopf holonomy takes values in the fiber subgroup $U(1) \subset SU(2)$, whereas the geometric phase represents the corresponding physical rotation in $SO(3)$. Thus, the weight-two representation $\rho_2(u) = u^2$ reflects precisely this double-covering structure.

4.3 Geometric phase and the geodesic curvature

Let us denote the geometric phase for an arbitrary time $t \in [0, 1]$ of the motion (M) by $\Delta_g(t)$, such that $\Delta_g(0) = 0$ and $\Delta_g(1) = \Delta_g$, namely,

$$\Delta_g(t) = - \int_0^t \cos \beta(t) \dot{\theta}(t) dt. \quad (4.41)$$

This value $\Delta_g(t)$ has a nice geometrical interpretation in terms of the motion on S^2 (Prop. 4.10). After preparing some notions of local frame and the geodesic curvature in the subsequent paragraphs, we shall elucidate this point. In the following discussion, the formulas are understood on each smooth segment of the Gauss curve away from the poles; when necessary, they are extended by the regularization introduced in § 2.

Local moving frame

At $\mathbf{g}(\theta, \beta) \in S^2$, let us introduce the local orthonormal frame by

$$\begin{aligned} \mathbf{e}_1 &= \frac{\partial \mathbf{g}}{\partial \theta} / \left| \frac{\partial \mathbf{g}}{\partial \theta} \right| = \begin{pmatrix} -\sin \theta \\ \cos \theta \\ 0 \end{pmatrix}, & \mathbf{e}_2 &= -\frac{\partial \mathbf{g}}{\partial \beta} / \left| \frac{\partial \mathbf{g}}{\partial \beta} \right| = \begin{pmatrix} -\cos \beta \cos \theta \\ -\cos \beta \sin \theta \\ \sin \beta \end{pmatrix}, \\ \mathbf{e}_3 &= \mathbf{e}_1 \times \mathbf{e}_2 = \begin{pmatrix} \sin \beta \cos \theta \\ \sin \beta \sin \theta \\ \cos \beta \end{pmatrix}, \end{aligned} \quad (4.42)$$

where $\mathbf{g}(\theta, \beta) \in S^2$ is the Gauss vector (2.5). Note that $\mathbf{e}_3 = \mathbf{g}$. By the definition, they satisfy

$$\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}, \quad \mathbf{e}_i \times \mathbf{e}_j = \sum_{k=1,2,3} \epsilon_{ijk} \mathbf{e}_k \quad (i, j = 1, 2, 3), \quad (4.43)$$

where ϵ_{ijk} is the complete antisymmetric tensor normalized as $\epsilon_{123} = 1$. Noting that the vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ are always oriented to the *east* and *north*, respectively, and $\mathbf{e}_3$ is to the *sky*. In particular, $\mathbf{e}_1$ and $\mathbf{e}_2$ span the tangent space $T_{\mathbf{g}}S^2$.

The *connection matrix* (ω_{ij}) is defined by

$$\begin{pmatrix} d\mathbf{e}_1 \\ d\mathbf{e}_2 \\ d\mathbf{e}_3 \end{pmatrix} = \begin{pmatrix} \omega_{11} & \omega_{12} & \omega_{13} \\ \omega_{21} & \omega_{22} & \omega_{23} \\ \omega_{31} & \omega_{32} & \omega_{33} \end{pmatrix} \begin{pmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \\ \mathbf{e}_3 \end{pmatrix}. \quad (4.44)$$

By the orthonormal condition (4.43), it immediately follows that ω is anti-symmetric

$$\omega_{ij} = -\omega_{ji} = d\mathbf{e}_i \cdot \mathbf{e}_j \quad \text{for} \quad i, j = 1, 2, 3. \quad (4.45)$$

It is explicitly calculated as

$$\begin{pmatrix} \omega_{11} & \omega_{12} & \omega_{13} \\ \omega_{21} & \omega_{22} & \omega_{23} \\ \omega_{31} & \omega_{32} & \omega_{33} \end{pmatrix} = \begin{pmatrix} 0 & \cos \beta d\theta & -\sin \beta d\theta \\ -\cos \beta d\theta & 0 & d\beta \\ \sin \beta d\theta & -d\beta & 0 \end{pmatrix}. \quad (4.46)$$

In particular, we see that $\omega_{12} = \cos \beta d\theta$.

Reparameterization from t to s

To describe the motion of disc B, it is convenient to adopt the *length* of the curve γ rather than the *time* parameter $t \in [0, 1]$. We shall express the length parameter by $s \in [0, L(\gamma)]$,

where $L(\gamma)$ is the length of $\gamma \subset S^2$, and regard $\mathbf{g}(s)$ as a function of the length s rather than the time t . The orientation of the curve parametrized by s is induced from that of the time parameter t such that

$$\mathbf{g}(t)|_{t=0} = \mathbf{g}(s)|_{s=0} \quad \text{and} \quad \mathbf{g}(t)|_{t=1} = \mathbf{g}(s)|_{s=L(\gamma)}. \quad (4.47)$$

On an open neighborhood for a fixed $t \in (0, 1)$, where $\mathbf{g}(t)$ is differentiable and $\mathbf{g}'(t) \neq 0$, the length parameter s is related to the time parameter t via the reparameterization

$$s : [0, 1] \rightarrow [0, L(\gamma)], \quad t \mapsto s(t) \quad \text{such that} \quad |\mathbf{g}'(s)| = |\mathbf{g}'(t)| \frac{dt}{ds} = 1. \quad (4.48)$$

By the definition, the speed of $\mathbf{g}(s)$ is normalized as 1 for the parameter s . We also see that ds^2 is the line element of S^2 . By (4.48) and (4.46), we have

$$ds = |\mathbf{g}'(t)| dt, \quad \mathbf{g}'(t) = \frac{d\mathbf{e}_3}{dt} = \sin \beta \frac{d\theta}{dt} \mathbf{e}_1 - \frac{d\beta}{dt} \mathbf{e}_2. \quad (4.49)$$

Therefore, we obtain

$$ds^2 = \sin^2 \beta d\theta^2 + d\beta^2. \quad (4.50)$$

This is nothing but the line element of S^2 in terms of the local coordinates (θ, β) . In the subsequent argument, we reserve s for the length parameter.

Geodesic curvature

Due to $|\mathbf{g}'(s)| = 1$ and $\mathbf{g}'(s) \in T_{\mathbf{g}(s)}S^2 = \text{span}\{\mathbf{e}_1, \mathbf{e}_2\}$, we are able to express the tangent vector $\mathbf{g}'(s)$ by introducing an angle ϕ ,⁴

$$\mathbf{g}'(s) = \cos \phi \mathbf{e}_1 + \sin \phi \mathbf{e}_2, \quad \text{where} \quad \cos \phi = \sin \beta \frac{d\theta}{ds}, \quad \sin \phi = -\frac{d\beta}{ds}. \quad (4.51)$$

Figure 4 illustrates the geometric meaning of the angle ϕ . Replacing ϕ by $\phi + \pi/2$, we have the orthogonal vector defined by

$$\boldsymbol{\nu}(s) = -\sin \phi \mathbf{e}_1 + \cos \phi \mathbf{e}_2. \quad (4.52)$$

Note that they satisfy

$$|\mathbf{g}'(s)| = |\boldsymbol{\nu}(s)| = 1, \quad \mathbf{g}'(s) \cdot \boldsymbol{\nu}(s) = 0. \quad (4.53)$$

The relation between the local frames $\{\mathbf{e}_1, \mathbf{e}_2\}$ and $\{\mathbf{g}'(s), \boldsymbol{\nu}(s)\}$ is presented in Fig. 4. Differentiating the second relation, we get

$$\mathbf{g}''(s) \cdot \boldsymbol{\nu}(s) + \mathbf{g}'(s) \cdot \boldsymbol{\nu}'(s) = 0. \quad (4.54)$$

⁴In our previous work [8], the angle ϕ was expressed by the symbol φ , that is $\phi_{[\text{here}]} = \varphi_{[8]}$. While φ is reserved for the $U(1)$ fiber coordinate in this article.

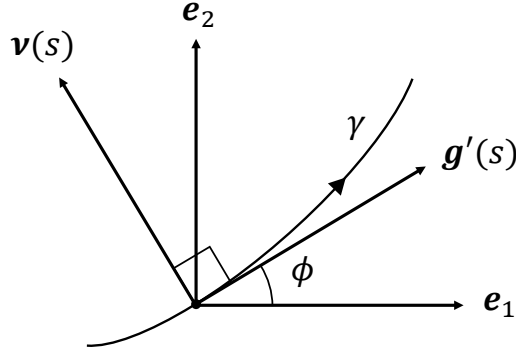


Figure 4: The moving frames $\{\mathbf{g}'(s), \boldsymbol{\nu}(s)\}$ and $\{\mathbf{e}_1, \mathbf{e}_2\}$ at $\mathbf{g}(s) \in S^2$. The vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ are oriented to east and north, respectively. ϕ is the angle between the local frames.

Definition 4.8 (Geodesic curvature). The *geodesic curvature* of $\gamma \subset S^2$ at $\mathbf{g}(s) \in \gamma$ is defined by

$$\kappa_g(s) = \mathbf{g}''(s) \cdot \boldsymbol{\nu}(s) = -\mathbf{g}'(s) \cdot \boldsymbol{\nu}'(s). \quad (4.55)$$

By plugging $\mathbf{g}'(s)$ in (4.51) and

$$\boldsymbol{\nu}'(s)ds = -\cos \phi(d\phi + \omega_{12})\mathbf{e}_1 - \sin \phi(d\phi + \omega_{12})\mathbf{e}_2 + (-\sin \phi \omega_{13} + \cos \phi \omega_{23})\mathbf{e}_3, \quad (4.56)$$

with the geodesic curvature (4.55), the geodesic curvature is explicitly written as

$$\kappa_g(s)ds = d\phi + \omega_{12} = d\phi + \cos \beta d\theta. \quad (4.57)$$

This relation also allows us to express the canonical connection of the Hopf fibration directly in terms of the geometry of the Gauss curve.

Lemma 4.9. *The canonical connection ω in (3.46) admits the following expression in terms of the local moving frame along the Gauss curve and its geodesic curvature:*

$$\omega = \frac{i}{2}(d\varphi - d\phi + \kappa_g(s)ds), \quad (4.58)$$

where φ is the Hopf angular coordinate in (3.45), ϕ is the angle defined in (4.51), and κ_g is the geodesic curvature defined in (4.55).

Proof. The desired expression immediately follows by substituting

$$\cos \beta d\theta = \kappa_g(s)ds - d\phi \quad (4.59)$$

for the canonical connection ω in (3.46). $\square$

On the other hand, the tangent vector d/dt of the horizontal lift of the Gauss curve γ is characterized by the condition $\omega(d/dt) = 0$, which is equivalent to (3.55) and written by

$$\dot{\varphi} - \dot{\phi} + \kappa_g(s) \frac{ds}{dt} = 0. \quad (4.60)$$

Integrating this with respect to t , we obtain the following proposition.

Proposition 4.10. *Let $\Delta_g(t)$ be the geometric phase for time $t \in [0, 1]$ defined in (4.41), $\phi(t)$ an angle given in (4.51), and $\kappa_g(s)$ the geodesic curvature of γ in (4.55). Then we have*

$$\Delta_g(t) = \phi(t) - \phi(0) - \int_0^{s(t)} \kappa_g(s) ds. \quad (4.61)$$

Proof. By the expression of ω in (3.46) and the horizontal condition (4.60), we have

$$\begin{aligned} \Delta_g(t) &= - \int_0^t \cos \beta(t) \dot{\theta}(t) dt = \int_0^t \dot{\varphi}(t) dt \\ &= \int_0^t (\dot{\phi} - \kappa_g(s) \frac{ds}{dt}) dt = \phi(t) - \phi(0) - \int_{s(0)}^{s(t)} \kappa_g(s) ds. \end{aligned} \quad (4.62)$$

As $s(0) = 0$ by the definition, we obtain the desired relation. $\square$

Corollary 4.11. *Suppose that the Gauss curve γ is simple. Then, the geometric phase Δ_g is expressed as*

$$\begin{aligned} \Delta_g &= 2\pi(1 - I_+) - \int_0^{L(\gamma)} \kappa_g(s) ds \\ &= 2\pi(I_- - 1) - \int_0^{L(\gamma)} \kappa_g(s) ds \\ &= \pi(I_- - I_+) - \int_0^{L(\gamma)} \kappa_g(s) ds, \end{aligned} \quad (4.63)$$

where $I_{\pm} \in \{0, 1, 2\}$ are the topological indices, namely, the numbers of the corresponding poles enclosed by the regularized curve $\gamma(\epsilon)$ in (2.10) on its left-hand side.

Proof. We only show the first relation (4.63) since the remaining equations follow from $I_+ + I_- = 2$. By Prop. 4.10 and setting $t = 1$, we get

$$\Delta_g = \phi(1) - \phi(0) - \int_0^{L(\gamma)} \kappa_g(s) ds, \quad (4.64)$$

since $s(1) = L(\gamma)$. After applying the same regularization as in the definition of $I_{\pm}$ (and smoothing corners if necessary), the total turning of the tangent direction is

$$\phi(1) - \phi(0) = 2\pi(1 - I_+). \quad (4.65)$$

Hence, this completes the proof. $\square$

Remark 4.12. Finally, we comment on the geometric interpretation of Prop. 4.10. The angle ϕ introduced in (4.51) measures the direction of the tangent vector $\mathbf{g}'(s)$ relative to the local frame $\mathbf{e}_1$, which may be regarded as the needle of a *compass*. Hence, $\phi(1) - \phi(0)$ measures the net rotation of disc B relative to the compass. On the other hand, the integral of the geodesic curvature κ_g measures the total intrinsic turning of the Gauss curve. Through the horizontal condition $\omega = 0$ in (4.60), these quantities are related to the Hopf angular coordinate φ . Since the geometric phase is related to φ by (4.41), Prop. 4.10 expresses $\Delta_g(t)$ as a combination of the rotation relative to the compass and the integrated geodesic curvature. We expect that this viewpoint will be useful in studying the Foucault pendulum [13].⁵

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A Diffeomorphism from S^2 to $\mathbb{CP}^1$

To fix the convention of local coordinates of S^2 and $\mathbb{CP}^1$, we shall spell out a diffeomorphism of S^2 and $\mathbb{CP}^1$.

Two-sphere S^2 is defined by

$$S^2 = \{ (a, b, c) \in \mathbb{R}^3 \mid a^2 + b^2 + c^2 = 1 \} \subset \mathbb{R}^3, \quad (\text{A.1})$$

with an atlas $\{(V_1, g_1), (V_2, g_2)\}$, where the local charts are given by the stereographic projection, namely,

$$\begin{aligned} g_1 : V_1 = \{ (a, b, c) \in S^2 \mid c \neq -1 \} &\rightarrow \mathbb{C}, & (a, b, c) &\mapsto \frac{a - ib}{1 + c}, \\ g_2 : V_2 = \{ (a, b, c) \in S^2 \mid c \neq 1 \} &\rightarrow \mathbb{C}, & (a, b, c) &\mapsto \frac{a + ib}{1 - c}. \end{aligned} \quad (\text{A.2})$$

⁵Geometric approaches to the Foucault pendulum are discussed in [14, 15, 16, 17, 18, 19, 20, 21, 22].

On $g_1(V_1 \cap V_2) = g_2(V_1 \cap V_2) = \mathbb{C}^*$, the transition map is

$$g_2 \circ g_1^{-1} : \mathbb{C}^* \rightarrow \mathbb{C}^*, \quad z \mapsto \frac{1}{z}. \quad (\text{A.3})$$

In terms of the angular coordinates β, θ ($0 \leq \beta \leq \pi, 0 \leq \theta < 2\pi$),

$$a = \sin \beta \cos \theta, \quad b = \sin \beta \sin \theta, \quad c = \cos \beta, \quad (\text{A.4})$$

it can be written as

$$\begin{aligned} g_1((a, b, c)) &= \frac{e^{-i\theta} \sin \beta}{1 + \cos \beta} = e^{-i\theta} \tan \frac{\beta}{2}, \\ g_2((a, b, c)) &= \frac{e^{i\theta} \sin \beta}{1 - \cos \beta} = e^{i\theta} \cot \frac{\beta}{2}. \end{aligned} \quad (\text{A.5})$$

On the other hand, $\mathbb{CP}^1$ is given by

$$\mathbb{CP}^1 = \{ [z_1, z_2] \mid (z_1, z_2) \in \mathbb{C}^2 \setminus \{(0, 0)\} \}, \quad (\text{A.6})$$

where the equivalence relation $(z_1, z_2) \sim (w_1, w_2)$ is defined by

$$(z_1, z_2) = (\lambda w_1, \lambda w_2) \quad \text{with} \quad \lambda \in \mathbb{C}^\times. \quad (\text{A.7})$$

An atlas $\{(U_1, f_1), (U_2, f_2)\}$ is defined by

$$\begin{aligned} f_1 : U_1 &= \{ [z_1, z_2] \mid z_1 \neq 0 \} \rightarrow \mathbb{C}, \quad [z_1, z_2] \mapsto \frac{z_2}{z_1}, \\ f_2 : U_2 &= \{ [z_1, z_2] \mid z_2 \neq 0 \} \rightarrow \mathbb{C}, \quad [z_1, z_2] \mapsto \frac{z_1}{z_2}. \end{aligned} \quad (\text{A.8})$$

On $f_1(U_1 \cap U_2) = f_2(U_1 \cap U_2) = \mathbb{C}^*$, the transition map reads

$$f_2 \circ f_1^{-1} : \mathbb{C}^* \rightarrow \mathbb{C}^*, \quad z \mapsto \frac{1}{z}. \quad (\text{A.9})$$

Proposition A.1. *A map $h : S^2 \rightarrow \mathbb{CP}^1$ defined by*

$$\begin{aligned} h : V_1 &\rightarrow U_1 \subset \mathbb{CP}^1, \quad (a, b, c) \mapsto \left[1 : \frac{a - ib}{1 + c} \right], \\ h : V_2 &\rightarrow U_2 \subset \mathbb{CP}^1, \quad (a, b, c) \mapsto \left[\frac{a + ib}{1 - c} : 1 \right], \end{aligned} \quad (\text{A.10})$$

is a diffeomorphism from S^2 to $\mathbb{CP}^1$.

Proof. The map h is smooth since for $i, j = 1, 2$ it holds that

$$\begin{aligned} f_i \circ h \circ g_i^{-1} : \mathbb{C} &\rightarrow \mathbb{C}, \quad z \mapsto z, \\ f_j \circ h \circ g_i^{-1} : \mathbb{C}^* &\rightarrow \mathbb{C}^*, \quad z \mapsto \frac{1}{z}, \quad (i \neq j). \end{aligned} \quad (\text{A.11})$$

The inverse map $h^{-1} : \mathbb{CP}^1 \rightarrow S^2$ given by

$$\begin{aligned} h^{-1} : U_1 &\rightarrow V_1 \subset S^2, \quad [1 : z] \mapsto \left(\frac{2\operatorname{Re}z}{1 + |z|^2}, \frac{-2\operatorname{Im}z}{1 + |z|^2}, \frac{1 - |z|^2}{1 + |z|^2} \right), \\ h^{-1} : U_2 &\rightarrow V_2 \subset S^2, \quad [z : 1] \mapsto \left(\frac{2\operatorname{Re}z}{1 + |z|^2}, \frac{2\operatorname{Im}z}{1 + |z|^2}, \frac{|z|^2 - 1}{1 + |z|^2} \right) \end{aligned} \quad (\text{A.12})$$

is also smooth. Hence, the map $h : S^2 \rightarrow \mathbb{CP}^1$ in (A.10) defines a diffeomorphism from S^2 to $\mathbb{CP}^1$. This completes the proof. $\square$

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